(a) Action-aligned dataset construction

1 Filter & normalize


DROID


RH20T

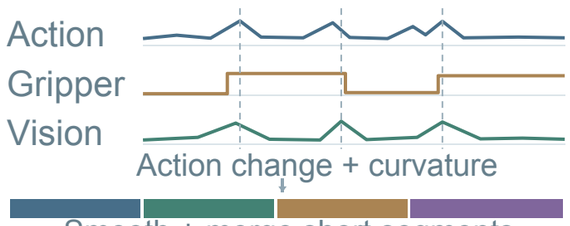

AgiBot-World


MolmoAct

✓ Text
✓ Images
✓ Actions

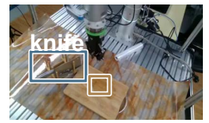
RGB + 32-D state / action

2 Candidate boundaries

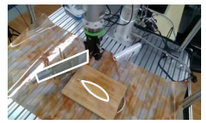


3 Interaction evidence

GroundingDINO-B



SAM 3.1



Tracks + motion
Gripper + contact
Active object / target / phase

4 Segment refinement



Phase + roles + space + actions

Action-aligned segment records

5 Grade-conditioned rewrites

Qwen3.8-27B

	A	T	V
Low	✓	✓	—
Mid	✓	✓	✓
High	—	✓	✓

A Atomic motion
Move downward.

T Local task
Approach the knife.

V Visual prompt
Approach the object in the red box.

6 Sampling metadata

Task Subtask
Phase Approach
Family A / T / V
Scope **Mid**

w1 w2 w3
Weights

7 Multimodal review

GPT-5.6-luna



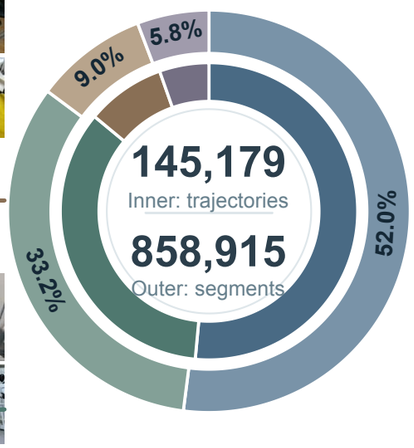
✓ Language ✓ Visual target ✓ Action scope

⊕ Keep / expand supported labels
⊗ Disable inconsistent labels

(b) Dataset overview


RH20T


DROID



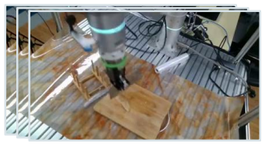
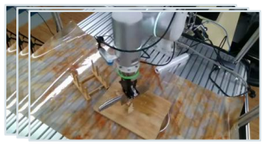
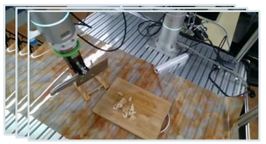
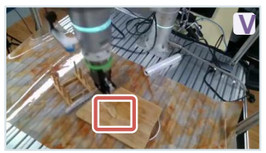

AgiBot-World


MolmoAct

(c) Instruction examples

Task: Slice the lotus root with a knife.

A Atomic motion T Task rewrite V Visual prompt

S1 Approach (Mid)	S2 Grasp (High)	S3 Transfer / align (Mid)	S4 Slice (High)	S5 Withdraw (Mid)
				
A Move downward.	A N/A	A Keep the gripper closed and move right.	A N/A	A Keep the gripper closed and move right.
T Approach the knife.	T Grasp the knife.	T Align the knife above the root.	T Slice the lotus root.	T Withdraw the held knife.
				
Approach the object in the red box.	Grasp the object in the red box.	Move the object toward the red point.	Slice the object in the red region.	Move the object away from the red box.